RESEARCH REPORT

Imbalance Issues in the CHB-MIT Scalp EEG Database

Exploratory Data Analysis | Epileptic Seizure Detection | Pediatric Neurology

Introduction

The CHB-MIT Scalp EEG Database is a publicly available benchmark dataset widely used for epileptic seizure detection research. Developed through collaboration between Children's Hospital Boston (CHB) and MIT, and hosted on PhysioNet, the dataset contains multi-channel scalp EEG recordings from pediatric epilepsy patients with annotated seizure events in European Data Format (EDF).

During the Exploratory Data Analysis (EDA) phase, several critical imbalance-related issues were identified. These characteristics directly influence the learning capability, reliability, and generalization of deep learning models trained on the dataset.

Dataset Overview

Parameter	Value
Total Patients	24
Total EDF Files	686
Seizure Files	141
Total Files	851
Age Range	1.5–22 yrs

Identified Imbalance Issues

01 Class Imbalance

Seizure recordings represent only a minor fraction of the total EDF files (141 of 686), resulting in a heavily skewed class distribution. This disparity makes it difficult for models to learn seizure-specific patterns without bias toward the majority non-seizure class.

- Seizure files: 141 / 686 total
- Models risk predicting non-seizure by default

02 Patient-Level Imbalance

EEG recording counts vary dramatically across patients. Highly represented patients dominate training data, potentially overshadowing EEG characteristics from underrepresented subjects and reducing model generalizability across individuals.

- chb01: 42 files | chb15: 40 files
- chb23: only 9 files

03 Seizure Frequency Imbalance

Seizure occurrence rates differ substantially across patients. High-ratio patients (chb24: 0.545, chb12: 0.542) flood the seizure class, while low-ratio patients (chb04: 0.071) contribute minimally, creating inconsistency in seizure representation.

- Highest ratio: 0.545 (chb24)
- Lowest ratio: 0.071 (chb04)

04 Demographic Imbalance

With subjects aged 1.5–22 years and a female-dominant cohort, the dataset provides limited adult EEG representation. This skews trained models toward pediatric signal patterns and reduces demographic coverage, restricting applicability to broader patient populations.

- Pediatric-dominant (age 1.5–22 years)
- Female patients outnumber male patients

05 Data Leakage Risk

Multiple EEG recordings per patient create a leakage risk when the same subject appears in both training and test sets. Models may memorize patient-specific EEG fingerprints rather than generalizable seizure features, inflating performance metrics artificially.

- Patient-wise split is mandatory for valid evaluation
- Cross-patient generalization must be verified

06 Recording Variability

EDF recordings differ in duration, channel configuration, and seizure event distribution. Some files contain multiple seizures while others contain exclusively normal EEG activity. This inconsistency increases dataset complexity and creates challenges in developing uniform preprocessing and segmentation pipelines.

Conclusion & Research Implications

The EDA phase reveals that the CHB-MIT dataset carries six distinct imbalance challenges — class imbalance, patient-level imbalance, seizure frequency imbalance, demographic imbalance, data leakage risk, and recording variability. Each issue poses a unique threat to model reliability and fairness. Addressing these challenges through patient-wise data splitting, oversampling strategies, and demographic-aware evaluation frameworks is essential before applying deep learning techniques for seizure detection.

#	Issue	Key Impact
1	**Class Imbalance**	Model bias toward non-seizure predictions
2	**Patient-Level Imbalance**	Unequal EEG characteristic representation
3	**Data Leakage Risk**	Artificially inflated evaluation performance
4	**Seizure Frequency Imbalance**	Inconsistent seizure representation across subjects
5	**Demographic Imbalance**	Limited generalization beyond pediatric population